MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8122

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Any
Weight: 3.5 ounces nominal (99.2 grams)

ABSOLUTE RATINGS: F1 = 500 MHz (see note 1)

Parameter:	Ef	Eb	Ec2	Ec1	Ehk	Ib	Ic1	Pg2
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	W
Maximum:								
Linear rf power amp-SSB-SCS:	13.5 ±10%	2,200	400	-100	150	300 (Note 2)	100	8
Anode mod rf power amp C telep:	13.5 ±10%	1,800	400	-100	150	250	100	5
RF power amp and osc-C teleg:	13.5 ±10%	2,200	400	-100	150	300	100	8
TEST CONDITIONS:	13.5	700	250	-100	---	185	---	---

ABSOLUTE RATINGS: F1 = 500 MHz (see note 1)

Parameter:	Pp	tk	T (seal and anode core)	Cooling
Unit:	W	sec(min)	°C	Note 3
Maximum:				
Linear rf power amp-SSB-SCS:	400	60	250	---
Anode mod rf power amp C telep:	280	60	250	---
RF power amp and osc-C teleg:	400	60	250	---
TEST CONDITIONS:	---	120	---	Note 12

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1201	Short and discontinuity detection	13		0.65	II	---	---	---	---
1266	Total grid current	4,16		0.65	II	Ic1	---	-25	µAdc
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec1	-8	-19	Vdc
1261	Electrode voltage (2) (grid)	-	Eb = 2,000 Vdc; Ec1/Ib = 5 mAdc	0.65	II	Ec1	---	-47	Vdc
1256	Electrode current (screen)	-		0.65	II	Ic2	-7	+6	mAdc
1231	Pulsing emission sinusoid	4,5	eb = ec2 = ec1 = 200 v; tp = 2.0 µs (min), 10.0 µs (max); prr = 60; t = 60	0.65	II	is	13	---	a
1301	Heater current	-		0.65	II	If	1.15	1.45	A
---	Standoff voltage	14		0.65	II	SV	1.5	---	kVdc
	Quality conformance inspection, part 2								
1266	Primary grid emission (control)	15	eg1/Pg1 = 4.0 W; t = 2 minutes (max); anode and screen grid grounded	---	---	Isg1	---	50	µAdc
1266	Primary grid emission (screen)	15	eg2/Pg2 = 8.0 W; t = 2 minutes (max); anode and control grid grounded	---	---	Isg2	---	50	µAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
---	Zero-bias anode current	6	Eb = 450 Vdc; Ec2 = 400 Vdc; Ec1 = -100 Vdc; tp = 4,500 to 5,000 µs; prr = 10 to 12	---	---	ib	1.0	1.8	a
2214	RF useful power output	7	F = 470 MHz; Ef = 12.5; Eb = 2,000 Vdc; Ec1 = -30 ±2 Vdc; Ec2 = 400 Vdc (max); Ic1 = 30 mAdc; Ec2/Ib = 300 mAdc; Pd = 8 W; T = 225° ±25°C	---	---	Po	275	---	W
1331	Direct-interelectrode capacitance	-	Measured with shielded adapter	---	---	Cg1-p	---	0.15	pF
						Cg1-k	14.0	18.0	pF
						Cp-k	0.004	0.016	pF
						Cg1-g2	20.0	26.5	pF
						Cg2-p	6.3	7.7	pF
						Cg2-k	2.1	3.3	pF
						Ch-k	2.5	4.1	pF
1366	Interelement leakage resistance (cold)	8	Supply voltage = 200 Vdc; Rs = 1.0 Meg	---	---	Rg1g2	50	---	Meg
						Rg1k	50	---	Meg
						Rg2p	50	---	Meg
1336	Heater-cathode leakage	-	Ehk = +150 Vdc	---	---	Ihk	---	150	µAdc
			Ehk = -150 Vdc			Ihk	---	150	µAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
---	Life test	-	Group D; rf useful output power; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
2214	RF useful power output	-		---	---	Po	250	---	W
1266	Primary grid emission (control)	-		---	---	Isg1	---	-500	µAdc
1266	Primary grid emission (screen)	-		---	---	Isg2	---	-500	µAdc
1336	Heater-cathode leakage	-	Ehk = +150 Vdc Ehk = -150 Vdc	---	---	Ihk Ihk	--- ---	150 150	µAdc µAdc
	Periodic-check tests	11							
1032	Vibration, mechanical	9	No voltages; frequency 10 to 2,000 Hz; Accel = 15 G	---	---	---	---	---	---
1042	Shock, specified pulse	10	Test condition A	---	---	---	---	---	---
---	Vibration and shock test end points:								
1266	Total grid current	-		---	---	Ic1	---	-25	µAdc
1261	Electrode voltage (1) (grid)	-		---	---	Ec1	-8	-19	Vdc
1256	Electrode current (screen)	-		---	---	Ic2	-7	+6	mAdc

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NOTES:

1. Because the cathode is subjected to considerable back bombardment as the frequency is increased with resultant increase in temperature, the heater voltage should be reduced depending on operating conditions and frequency to prevent overheating the cathode and resultant short life.

2. The maximum rating for a signal having a minimum peak-to-average power ratio less than 2, as obtained in "single-tone" operation, is 300 mAdc. During short periods of circuit adjustment under "single-tone" conditions, the average anode current may be as high as 450 mAdc.

3. In all applications, sufficient cooling shall be supplied to maintain the anode conduction cylinder and the base seals within the absolute maximum temperature rating of 250°C. In applications where reliable performance and long life are factors, operation at temperatures below the absolute maximum rating is normally beneficial.

4. This test shall be the first test performed after the holding period.

5. The voltage-pulse amplitude is 200 V peak. After 1 minute at this value, the current-pulse amplitude will not be less than the value specified.

6. Drive grid positive to zero volts with a square pulse. Zero bias anode current is the peak anode current at the start of the pulse.

7. Use grid driven coaxial amplifier cavity equipment.

8. The tube shall be cooled (room ambient temperature) 30 minutes before performing this test. The minimum resistance between any two electrodes as measured with a 200-volt Megger-type ohmmeter, or equivalent, shall be as specified.

9. Vibrate along each of the three mutually perpendicular axes (X, Y, and Z), see figure 1.

10. The tube shall be subjected to 18 impact shocks (see figure 1 for orientation of axes) as follows: Three shocks in each of three mutually perpendicular axes in the following sequence: Y+, Y-, X+, X-, Z+, and Z-. This is not a destructive test.

11. This test shall be performed every 3 months. Sampling shall be made as follows using a separate sample for each test:

Lot size	Size	Initial sample		Second sample size	Combined sample		
		Accept No. of failures	Reject No. of failures		Size	Accept	Reject
1-99	1	0	---	2	3	1	2
100-199	2	0	---	4	6	2	3
200-299	3	0	---	6	9	3	4
300-399	4	0	---	8	12	4	5
400-599	8	1	7	12	20	7	8
600-799	12	1	8	20	32	11	12
800-999	20	2	11	20	40	13	14

For lots larger than 1,000 tubes per month a reduced sampling plan shall be used as shown below, providing a process average has been established on the preceding 192 tubes.

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NOTES: -Continued

Process average %	Size	Initial sample		Second sample size	Combined sample		
		Accept No. of failures	Reject No. of failures		Size	Accept	Reject
0-5.00	4	0	---	16	20	7	8
5.01-15.00	12	1	8	20	32	11	12
15.01-20.00	20	2	11	20	40	13	14

In case of a sample failure, that specific test shall become a quality conformance inspection, part 2 test, AQL 6.5 percent, inspection level S3. After three consecutive successful submissions, the test shall revert to a quarterly periodic basis.

12. Sufficient airflow shall be supplied to limit all seals and anode core temperature to 250°C (maximum).

13. The hexaphase short and discontinuity system may be used as an alternate test to check for tube continuity and shorts. Continuity is indicated by a glow on one electrode of the neon indicator lamp connected to the tube element. A short circuit is indicated by a double glow on both electrodes of the neon lamps connected to the shorted tube elements.

14. The standoff voltage circuit as shown on figure 2, or equivalent may be used. The supply's internal impedance limits the short circuit current to about 1.5 mAdc. The voltmeter movement should be approximately 1.0 mAdc. Connect the positive (+) side of the supply to the screen grid of the tube under test as shown on figure 2. Standoff voltage is defined as the maximum voltmeter reading obtained at the point the voltage between g1 and g2 breaks down. A characteristic drop in the voltmeter reading is obtained at this point. The limit value of standoff voltage may be preset on the meter before inserting the tube. If the voltmeter does not rise to the preset value, the tube is defective.

15. This test shall be performed before the rf useful power output (Po) test. Seal temperature range setting under the application of air cooling and the specified applied electrode powder should be as follows:

(a) For primary grid emission (control) - G1 seal temperature = 120° to 150°C.
(b) For primary grid emission (screen) - G2 seal temperature = 120° to 150°C.

16. The anode core temperature shall be between 230° and 250°C during this test.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - ME, MU
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

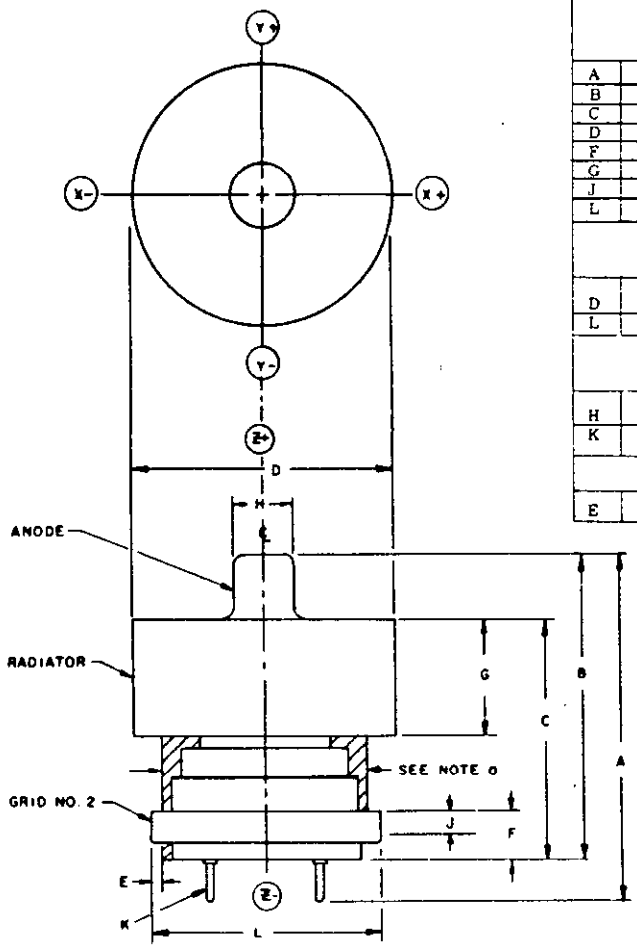
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2812)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		NOTES
	Minimum	Maximum	
Quality conformance inspection, part 1			d
A		2.26 (57.40)	
B	1.855 (47.12)	1.985 (50.42)	
C	1.470 (37.34)	1.560 (39.62)	
D	1.610 (40.89)	1.640 (41.66)	
F	.280 (7.11)	.320 (8.13)	
G	.710 (18.03)	.790 (20.07)	
J	.150 (3.81)		
L	1.416 (35.97)	1.436 (36.47)	
Quality conformance inspection, part 2			
D	---	---	b, c
L	---	---	b, c
Quality conformance inspection, part 3 (periodic check)			
H	Cap: C1-1 (EIA)		
K	Base: E11-81 (EIA)		b, c
Reference dimension			
E	.080 (2.03)		



Pin connections

1 - Cathode
2 - Grid No. 2
3 - Grid No. 1
4 - Cathode
5 - Heater
6 - Heater
7 - Grid No. 2
8 - Grid No. 1
9 - Cathode
10 - Grid No. 2
11 - Grid No. 1

NOTES:

a. Keep all stippled regions clear. Do not allow contacts or circuit components to protrude into these annular volumes.

b. The diameters of the radiator, grid No. 2 terminal contact surface, and the pin circle to be concentric within the following values of maximum full indicator reading:

Radiator to grid No. 2
Terminal contact surface---------- .030 (.76 mm) max.
Radiator to pin circle---------- .040 (1.02 mm) max.
Grid No. 2 terminal contact
Surface to pin circle---------- .030 (.76 mm) max.

c. The full indicator reading is the maximum deviation in radial position of a surface when the tube is completely rotated about the center of the reference surface. It is a measure of the total effect of run-out and ellipticity.

d. The AQL for dimensions listed under quality conformance inspection, part 1 shall be 1.0 percent, inspection level II.

FIGURE 1. Outline drawing of electron tube type 8122.

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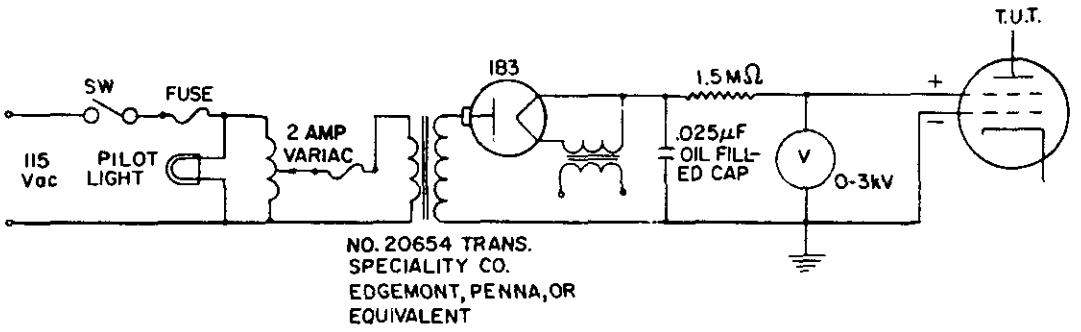


FIGURE 2. Standoff voltage circuit (Grid 1 to Grid 2).

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